

DAT560 Multimodal Fake-News Detection on FakeTT

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ABSTRACT

We study multimodal fake-news detection on FakeTT, a benchmark of 242 short TikTok videos with an inverted class distribution between training, validation and held-out test split. We evaluate two systems: System 1, a TF-IDF and Logistic Regression text baseline, and System 2, a prompted multimodal-LLM verifier that consumes the post text, four sampled video frames, and a balanced few-shot exemplar set drawn from the training split. On top of the ablation grid required by the DAT560 brief, we run a model-size scaling study at three scales, 8.8 B, 31.3 B, and 125.1 B parameters, and a multi-stage claim-extraction pipeline that distills each video’s spoken transcript and on-screen text into a structured list of factual claims for the verifier to consume. On the held-out test split, the best System 2 configuration is the 31 B verifier with three-shot frames and the extracted-claims block, reaching macro-F1 of 0.758, above the TF-IDF baseline at 0.627 and well above the trivial always-*real* floor at 0.403. The dominant pattern across the study is generalization rather than capability: the validation-best 125 B configuration loses 0.228 macro-F1 on test, while the simplest validation-defensible alternative nearly matches its validation score on the held-out split. We attribute this gap to the inverted class balance between validation and test, and document several other ablations transcripts, self-consistency, a two-stage describe-then-reason pipeline, that follow the same val-overfitting pattern.

1 INTRODUCTION

Context and Motivation. Short video platforms like TikTok and Instagram Reels have become primary channels for news consumption and social interaction, which makes the rapid spread of misleading content a pressing social problem. The FakeTT dataset provides a realistic, multimodal snapshot of this challenge by pairing short videos with textual metadata like descriptions and user captions, visual frames and human annotations for whether content is misleading or not. Single modality detectors like text only or vision only can miss important cross model cues, e.g. a convincing caption with contradicting visual evidence. So multimodal methods are a natural next step. However short video datasets introduce practical difficulties, metadata is noisy and informal, audio/ASR transcripts can be incomplete and dataset artifacts like repeated event identifiers may leak label information. Because of these issues, we emphasize reproducibility, clear ablation designs and conservative baseline choices to ensure any claimed improvements from multimodal LLMs are scientifically valid.

Research Problem. We address the binary classification task defined in the DAT560 brief: given a short video and its metadata, predict whether the video is misleading “fake” or not “real” and evaluate using macro-F1 as the headline metric. The core technical challenges are practical, the dataset is small with only 242 data points split into 139 for train, 29 for validation and 74 for test, class proportions shift between splits where in train 70% is fake vs in test where only 32% is fake. Also certain metadata fields such as “event” can unintentionally encode labels, we have observed label purity leakage in event identifiers. These properties make model selection and threshold tuning fragile when performed on the tiny validation set.

We build a reproducible TF-IDF + Logistic Regression baseline that carefully excludes leaking features and a modular MLLM verification pipeline that fuses textual, audio (ASR) and visual evidence using prompt engineering and ablation driven evaluation. Our emphasis is on robust experimental design so that any gains from prompt based multimodal methods generalize beyond the artifacts of the small validation split.

Related Work. Our work builds on several strands of prior research. Early studies showed that false information can spread differently than true information on social platforms, motivating automatic detection efforts [3]. Classical text classification techniques like TF-IDF with linear models remain reliable and interpretable baselines in misinformation research [2], which is why we reproduce these baseline methods carefully. Research on few-shot prompting and in-context learning demonstrated that large language models can learn tasks from examples without gradient updates [1], chain of thought and chain of verification approaches provide structured reasoning templates that improve LLM reliability on multi step inference[5].

Self consistency and ensemble prompting techniques further stabilize LLM outputs by aggregating multiple reasoning traces [4]. More recent vision language and multimodal LLM work has applied similar prompting and fusion strategies to content moderation and multimodal fact checking, but these methods require rigorous ablation and threshold control to avoid overfitting on small validation sets. Unlike much prior work that fixes a single model size or prompt, we systematically vary model scale and prompting templates and introduce a multi stage claims pipeline to probe where MLLMs truly add value.

Contribution Summary.

- A reproducible TF-IDF + Logistic Regression baseline for the required text-only system, with explicit handling of

event-leakage and experiments showing threshold instability when adding noisy ASR transcripts.

- A modular MLLM verification pipeline: config-driven, with a robust multi-strategy JSON parser, frame extraction + re-sizing, balanced-class few-shot exemplar selection, post-hoc threshold sweeping, and a self-consistency runner.
- All ablations required by the brief (frames, shot count, template, self-consistency, threshold tuning).
- A model-size scaling study spanning Qwen3-VL 8B, Gemma4 31B, and Qwen3.5 125B on val and test.
- A multi-stage claim-extraction pipeline: claims extracted by a 122B LLM, cached, and fed back to the verifier alongside frames and post text.
- On held-out test, System 2 outperforms System 1: best no-claims row reaches macro-F1 0.697 versus 0.627 for TF-IDF, and the best overall claims aware row reaches 0.758.

2 BACKGROUND

In this section we introduce the FakeTT dataset, our evaluation metric, the TF-IDF + Logistic Regression baseline and the prompting techniques used in the multimodal LLM system.

2.1 The FakeTT dataset

FakeTT is a small multimodal benchmark for fake news detection on short TikTok videos, with 242 posts split into train, validation and test sets. Each sample is paired with textual metadata fields such as "description", "user_description", "event", "user_certify" and "publish_time", together with the video itself and extracted visual evidence. The annotation rule is deliberately narrow, a post is labeled "fake" only if its specific factual claim is false, while stylistic sensationalism alone does not make the post fake. This distinction matters for later error analysis, because sensational wording can look suspicious even when the underlying claim is true.

A practical complication is that the class balance shifts strongly across splits, which makes model selection and threshold tuning sensitive to the choice of validation procedure. The table below summarizes the label distribution we use throughout the report.

Table 1: Dataset splits used in this study. The class balance is inverted between training/validation and the held-out test split, which has implications for any decision rule tuned on validation.

Split	#Samples	Fake	Real	Fake (%)	Real (%)
Train	139	98	41	70.5	29.5
Val	29	17	12	58.6	41.4
Test	74	24	50	32.4	67.6

Another important detail is that the "event" field is label leaky, across the splits event identifiers are highly label pure so using them in the main baseline can produce inflated performance that does not reflect genuine generalization. For that reason we exclude "event" from the primary TF-IDF baseline and treat it only as an ablation feature. We follow the FakeTT dataset description from the DAT560 brief and related project materials [6].

2.2 Macro-F1 as the evaluation metric

We report macro-F1 as the headline metric because it weights the two classes equally, regardless of class frequency. For the binary labels "fake" and "real", macro-F1 is

$$\text{Macro-F1} = \frac{1}{2} (F1_{\text{fake}} + F1_{\text{real}}).$$

This is preferable to accuracy in our setting because the class proportions differ substantially across train, validation and test. A model can achieve deceptively high accuracy by leaning toward the majority class on a particular split, yet still perform poorly on the minority class. In fact, a trivial always "real" predictor already attains 67.6% accuracy on test, while its macro-F1 is only 0.403. Throughout the report, we interpret macro-F1 together with the confusion matrix so that precision and recall trade offs remain visible.

2.3 Text classification with TF-IDF + Logistic Regression

Our supervised baseline represents each sample as a sparse TF-IDF vector built from the available text fields. TF-IDF scales a term by its frequency within a document and downweights terms that appear in many documents across the corpus. Informally for term t in document d ,

$$\text{tfidf}(t, d) = \text{tf}(t, d) \cdot \log \frac{N}{\text{df}(t)},$$

where N is the number of documents and $\text{df}(t)$ is the number of documents containing t . We use unigram and bigram features because short video metadata often contains useful multiword phrases that are lost with single words alone.

The TF-IDF vectors are then fed into Logistic Regression, a linear classifier that estimates $P(\text{fake} | x)$. To convert this probability into a label, we apply a decision threshold T : if $P(\text{fake}) \geq T$, the sample is predicted as "fake", otherwise it is predicted as "real". We tune T on the validation split by sweeping a grid from 0.05 to 0.95 and choosing the value that maximizes macro-F1. This makes threshold selection explicit and reproducible, which is especially important on a small validation set.

2.4 Multimodal LLMs

Multimodal LLMs combine text and image processing in a single transformer based system, typically by converting image inputs into tokens that are fused with the text token stream. In our project, the model is not asked to train a new classifier from scratch, instead it is prompted to act as a structured fact checker over the available evidence. We require the model to return machine parseable JSON containing a label, a confidence score and a short rationale so that downstream evaluation remains deterministic.

We compare three multimodal models served through Ollama on the UiS server, consisting of qwen3-vl:8b, gemma4:31b and qwen3.5:122b. This range lets us study whether prompt quality and model scale both matter or whether larger models mainly help because they can better integrate noisy multimodal cues.

2.5 Prompt-engineering techniques

Zero-shot prompting

Zero-shot prompting provides the model with only a task description (e.g., classify a news item as fake or real) without any labelled examples, relying entirely on the model’s pre-trained knowledge and generalization ability to perform the task.

Few-shot prompting

Few-shot prompting augments the task description with a small number k of labelled examples, typically balanced across classes (fake/real), allowing the model to infer the task pattern from demonstrations and adapt its predictions accordingly.

Role priming

Role priming frames the model with an explicit persona (e.g., “You are a professional fact-checker, be conservative about flagging fake”), which can steer its reasoning style, bias and decision thresholds by conditioning responses on the assumed role.

Chain-of-verification (CoV)

Chain-of-verification prompting instructs the model to decompose the task into steps: (i) extract key claims, (ii) verify each claim, (iii) identify inconsistencies, and (iv) produce a final classification, building on structured reasoning approaches such as chain-of-thought prompting → Chain-of-Thought Prompting Elicits Reasoning in Large Language Models [5].

Self-consistency

Self-consistency samples multiple model outputs at a non-zero temperature and aggregates them (e.g., via majority vote) to improve robustness and reasoning reliability, as proposed in Self-Consistency Improves Chain of Thought Reasoning in Language Models[4].

Confidence-threshold tuning

Confidence-threshold tuning is a post-hoc decision rule that accepts a “fake” classification only if the model’s self-reported confidence exceeds a predefined threshold T , aiming to reduce false positives by enforcing stricter acceptance criteria.

3 METHOD

In this section we describe the two systems under evaluation. System 1 is a required text-only supervised baseline. It uses a TF-IDF vectoriser followed by a logistic regression classifier. System 2 is a prompted multimodal large language model verifier. It consumes four sampled video frames, the post text, and a set of labelled few-shot examples, then returns a structured JSON verdict. We also describe a multi-stage claim extraction pipeline. This pipeline tests whether converting noisy transcripts and on-screen text into structured claim lists improves verification. We conclude with the reproducibility setup that connects all experiments.

3.1 Pipeline overview

Both systems share identical input and output contracts, while differing only in their internal processing. Each video is represented as a structured record that includes the raw dataset text fields, a reference to the video, and any precomputed artefacts required for the experiment. These artefacts include a speech transcript, on-screen text extracted with OCR, and optionally a list of extracted claims. Each system produces, for every sample, a verdict, a self-reported

confidence, and a short rationale. A shared evaluator converts these outputs into classification metrics. Keeping both interfaces fixed allows us to modify internal stages without affecting storage or comparison of results.

For each sample in System 2, we first concatenate the selected text fields into a single string. Next, we extract four frames spaced uniformly across the video and resize each so that the longest side does not exceed 384 pixels. We then construct the prompt by filling a template with the post text, an optional balanced few-shot block sampled from training data, and optionally a labelled set of extracted claims. Finally, we send both prompt and frames to a multimodal language model and parse the structured response into verdict, confidence, and rationale. Threshold sweeps and self-consistency voting are applied after inference. These procedures operate on stored outputs and do not require additional model calls.

Two design principles guide the pipeline. First, all experimental variation is controlled through configuration. Model choice, prompt template, number of examples, frame sampling, and decoding temperature are specified as settings rather than code changes. As a result, each ablation corresponds to a configuration file. Second, intermediate artefacts are computed once and reused. Frame extraction, Whisper transcription, OCR processing, and claim extraction are performed offline and cached. This design makes the scaling study in 4.4 practical. The same configuration can be evaluated across multiple models without repeating preprocessing.

3.2 System 1: TF-IDF and logistic regression

System 1 operates on text only. For each video, we combine two dataset fields, the post caption described as *description* and the user profile biography described as *user_description*. These are joined into a single document using a delimiter. We exclude the field *event*. This field is highly correlated with labels across splits, with approximately 94% of training events also appearing in the test split with the same majority label. Including it would introduce leakage rather than meaningful signal. We report an ablation that includes this field for completeness, but we do not treat it as a valid baseline.

The combined text is vectorised using TF-IDF with word unigrams and bigrams. The resulting sparse representation is passed to a logistic regression classifier. The classifier is trained on the training split. We select the decision threshold by maximising macro F1 on the validation split over a grid ranging from 0.05 to 0.95. This is the only parameter tuned on validation. Other parameters, including vocabulary size, regularisation strength, and n-gram range, remain at default values. This choice keeps comparison with System 2 simple and avoids overfitting given the small validation set. The model is then evaluated once on the test split. Predictions and metrics are computed a single time.

We also run two input-focused ablations. The event ablation adds the leakage-prone *event* field. This serves as an upper bound rather than a fair comparison. The transcript ablation adds speech transcripts generated by an automatic speech recognition system. 4.2 shows that this ablation is highly sensitive to threshold selection. The same model produces very different results depending on whether the threshold is tuned on validation or fixed at 0.5. This behaviour anticipates a similar validation overfitting pattern observed later in 4.5.

3.3 System 2: multimodal LLM verification pipeline

System 2 replaces the linear classifier with a prompted multimodal language model. Each input consists of three components: the post text, four sampled video frames, and a small set of labelled examples from the training split. Frames are sampled at uniform intervals across the video. Each frame is decoded and resized so that its longest side is at most 384 pixels. This resolution was chosen after observing that full resolution frames caused the smallest model to fail on many validation samples. The reduced size preserves enough visual detail while ensuring reliable generation.

The few-shot block is created once per run using a class-balanced sample of training videos. A slight majority is allocated to the *fake* class, which is the primary detection target. This block remains fixed for all samples in a run, ensuring that differences in output arise from the input sample rather than changing examples.

We evaluate three prompt templates. The first is a minimal template that defines the task, provides examples, and requests a verdict. The second adds role guidance and instructs the model to act conservatively when identifying fake content. The third follows a chain-of-verification approach. It decomposes the task into identifying claims, checking consistency with text and visuals, flagging unsupported claims, and producing a final classification. This decomposition aims to improve reasoning in smaller models.

All templates require a structured output that includes a binary verdict, a confidence score in $[0, 1]$, and a short rationale. This format supports both quantitative and qualitative analysis. The rationale is used for error inspection, while the confidence enables threshold-based evaluation.

Because outputs are not always well formed, we use a tolerant parsing strategy. We attempt direct parsing first, then remove surrounding formatting if necessary, and finally extract key fields using regular expressions. This approach achieved a full parse rate on validation.

Two post-processing analyses are applied. The threshold sweep evaluates predictions under varying confidence thresholds. A prediction is labelled as fake only if its confidence exceeds a threshold T , otherwise it is labelled as real. We evaluate all distinct confidence values produced by the model, along with boundary cases. The optimal threshold is selected directly from this set.

The self-consistency analysis generates multiple outputs for the same input at moderate temperature and selects the majority verdict. Ties are resolved in favour of the real class. To ensure independence, caching is disabled during this process.

3.4 Multi-stage claims pipeline

The claims pipeline explores a different hypothesis. It tests whether transforming transcripts and on-screen text into structured factual claims improves verification.

The pipeline consists of two stages. The first stage performs claim extraction and is executed once. For each video with available transcript or on-screen text, each source is processed independently using a large text language model. The model is prompted to produce a list of factual claims or return an empty list if none are present. A claim is defined as a concrete statement such as a date,

location, named entity, statistic, or attribution that can be verified externally. Opinions and general statements are excluded.

We use a large model for extraction to maximise quality and reuse the results across experiments. Approximately twelve percent of outputs were malformed and could not be parsed. These cases are treated as containing no claims, which is a conservative choice.

The second stage extends the verifier described in 3.3. Each input now includes a list of extracted claims with source labels that indicate whether they originate from speech or on-screen text. If no claims are available, the input explicitly states that no verifiable claims were extracted. Few-shot examples follow the same structure to maintain consistency.

All prompt templates are extended to support claim inputs. Each variant is evaluated using both mid-size and large multimodal models to determine whether structured claims provide greater benefit at specific scales.

4 EXPERIMENTAL EVALUATION

Experimental Evaluation outlines the dataset splits, model configurations, evaluation protocol, and metrics used to ensure fair and consistent comparison across all experiments.

4.1 Experimental setup

We use the same split definition and class distribution shown in Table 1 for all experiments in Sections 3 and 4. Validation is used only to select decoding strategy, prompt template, shot count, and any decision threshold; the test split is evaluated exactly once per locked configuration and is never used for tuning.

For System 2 we evaluate three multimodal-capable LLMs hosted on the University of Stavanger’s Ollama inference server and accessed remotely: an 8.8 B-parameter Qwen3-VL model, a 31.3 B-parameter Gemma 4 model, and a 125.1 B-parameter Qwen 3.5 mixture-of-experts model. Unless stated otherwise, generations use greedy decoding with temperature 0; the only experiments that use a non-zero temperature are the self-consistency analyses, which sample five generations at temperature 0.7 and majority-vote. A single random seed is used to draw the few-shot exemplars and any other stochastic input-construction step, so that two runs with the same configuration are deterministic up to server-side decoding noise.

The headline metric is macro-F1, the unweighted mean of per-class F1 over *fake* and *real*. We additionally report accuracy and per-class precision and recall in selected tables. When the model emits a response that cannot be parsed into a binary verdict, we fall back to *real*; this is the conservative choice given the trivial-classifier floor on test and ensures that every test sample contributes to the metrics.

4.2 System 1: TF-IDF baseline results

Table 2 reports System 1 on test. The primary baseline, TF-IDF over post text only, with a decision threshold tuned on validation – reaches a macro-F1 of 0.627, comfortably above the trivial always-*real* floor of 0.403. This is the number we treat as the System 1 reference throughout the rest of the report.

Adding the *event* field lifts macro-F1 to 0.704, but we report this row only as a known-leaky upper bound. Roughly 94% of the events

that appear in the training split are also present in test with the same majority label, so a classifier that learned event names would inherit a shortcut that has nothing to do with verifying the post’s claim. The lift therefore measures how much label information is encoded in the event field, not how good a fair text-only classifier could be.

The transcript ablation is the most interesting row in the table. With the decision threshold chosen on validation – the procedure mandated by the brief – adding speech transcripts collapses macro-F1 to 0.477, well below the no-transcript baseline. Holding the same vectoriser and classifier but fixing the threshold at the natural 0.5 cut-off instead recovers a macro-F1 of 0.704, now above the baseline. The classifier is unchanged; only the decision rule moved. The validation split has only twenty-nine samples, fewer than three quarters of which carry a non-empty transcript, and the macro-F1 surface as a function of threshold on so few samples is dominated by which side of a handful of borderline cases a particular threshold falls. The threshold that the validation procedure picks here does not transfer to test, where the class balance is inverted. We report the fixed-threshold row for transparency rather than as a recommendation: presenting the recovered 0.704 as the System 1 number would amount to retrospectively choosing a decision rule that happened to look good on test, which is exactly what the lock-from-val protocol is meant to prevent. The same dynamic recurs on the multimodal side in 4.5: configurations with the highest validation scores tend to be the ones whose advantage disappears on the held-out split.

Method	T	Acc.	Macro-F1	F1 _{fake}	P _{fake}	R _{fake}
Primary baseline (text only)	0.50	0.635	0.627	0.571	0.462	0.750
+ event field (leakage row)	0.55	0.743	0.704	0.596	0.609	0.583
+ transcript, val-tuned T	0.40	0.486	0.477	0.548	0.383	0.958
+ transcript, fixed T = 0.50	0.50	0.716	0.704	0.644	0.543	0.792

Table 2: System 1 (TF-IDF + Logistic Regression) results on the held-out test split. The primary baseline (bold) excludes the *event* field because it leaks the label across splits. Transcript rows are reported only as ablations: the validation-tuned threshold collapses on test, and the fixed-threshold variant violates the lock-from-val protocol.

4.3 System 2: ablation grid on validation

We now turn to System 2 and address each ablation required by the brief. Throughout this subsection the underlying model is the smallest of our three multimodal LLMs (8.8 B parameters); the same grid is repeated at larger scales in 4.4. Table 3 reports macro-F1 on the validation split across five axes: frames vs no frames, shot count, prompt template, self-consistency voting, and confidence-threshold tuning.

Including frames matters. On a zero-shot prompt, replacing post-text-only input with post text plus four sampled frames lifts macro-F1 from 0.515 to 0.589, an absolute gain of 0.074. Within the frame-using setting, varying the shot count traces a clear inverted-U: 0.589 at zero shots, 0.466 at one shot, 0.648 at three shots, and 0.570 at five shots. The peak at three shots matches the brief’s working assumption that small numbers of in-context examples help; the

Configuration	Macro-F1
<i>Frames vs no frames (zero-shot)</i>	
text only	0.515
+ 4 frames	0.589
<i>Shot count (frames, minimal template)</i>	
0-shot	0.589
1-shot	0.466
3-shot	0.648
5-shot	0.570
<i>Template (3-shot, frames)</i>	
minimal few-shot	0.648
role-priming	0.613
chain-of-verification	0.618
<i>Post-hoc analyses on the 3-shot frames row</i>	
self-consistency, voted (5x, T=0.7)	0.597
confidence-threshold sweep, best T	0.651

Table 3: System 2 ablation grid on the validation split for the smallest model (8.8 B parameters). All cells are macro-F1. Bold marks the best row in each block. The bottom block is post-hoc and consumes the 3-shot frames output without further model calls.

dip at one shot is striking and, as we will see in 4.4, vanishes at larger scales. For all subsequent comparisons in this subsection we hold the input at three shots with frames.

Among the three prompt templates, the minimal few-shot template wins at this scale. The role-priming variant that explicitly instructs the model to be conservative when flagging fakes scores 0.613, and the chain-of-verification variant that decomposes the task into list-check-flag-classify steps scores 0.618; both fall below the minimal template’s 0.648. The role-priming variant’s loss is concentrated in recall – the model becomes too cautious and misses some genuine fakes – while the chain-of-verification variant’s loss is dominated by the additional structure pushing the response away from clean JSON, costing two unparsed responses out of twenty-nine. At this model scale, more elaborate instructions hurt rather than help.

Sampling-based self-consistency does not help either. Five generations sampled at temperature 0.7 and majority-voted yield a voted macro-F1 of 0.597, below the greedy baseline of 0.648. The per-run scores are tightly clustered between 0.542 and 0.597, and the mistakes the runs make are highly correlated: seventeen of twenty-nine validation samples receive identical predictions across all five runs, and most of the voting errors come from this consistent-but-wrong group. Self-consistency is designed to average out stochastic decoding errors, but at this model scale the errors are systematic: the model carries a strong pro-*fake* prior that does not vary between samples drawn from the same prompt.

Confidence-threshold tuning, the final required ablation, is similarly close to a no-op on the best-performing configuration: sweeping the threshold at which the model’s self-reported *fake* verdict is accepted yields a macro-F1 of 0.651, an absolute lift of 0.003. The same rule applied to other configurations we explored – including

a two-stage describe-then-reason pipeline that we discuss alongside the test results in 4.5 – gives much larger lifts, of up to 0.23 on validation, confirming that threshold tuning is acting here as a bias-correction lever rather than as a calibration improvement. Taken together, the five ablations point in the same direction at this scale: minimal context, three balanced exemplars, and greedy decoding outperform every more elaborate variant we tried.

4.4 Model-size scaling on validation

We repeat the System 2 ablation grid at two larger scales, a 31.3 B Gemma 4 model and a 125.1 B Qwen 3.5 mixture-of-experts model. Table 4 reports the resulting macro-F1 grid on validation across all three model sizes. The headline row is the diagonal at three-shot frames, the configuration that won at 8 B: macro-F1 climbs from 0.648 to 0.780 going from 8 B to 31 B, and from 0.780 to 0.858 going from 31 B to 125 B. On a per-prompt basis, scale moves the metric more than any single ablation we tested in 4.3.

Configuration	8.8 B	31.3 B	125.1 B
<i>Shot count (frames, minimal template)</i>			
0-shot	0.589	0.664	0.748
1-shot	0.466	0.748	0.787
3-shot	0.648	0.780	0.858
5-shot	0.570	0.676	0.787
<i>Template (3-shot, frames)</i>			
minimal few-shot	0.648	0.780	0.858
role-priming	0.613	0.787	0.858
chain-of-verification	0.618	0.813	0.707
<i>Post-hoc analyses on 3-shot frames</i>			
self-consistency, voted	0.597	—	0.676
threshold sweep, best T	0.651	0.827	0.858

Table 4: Validation macro-F1 across three model sizes. Each row varies one axis of the System 2 grid; bold marks the best cell per row. The 31 B self-consistency cell could not be completed due to a transient remote-server error during the run.

The inverted-U over shot count survives at every scale, but its sharpness changes. At 8 B the curve is jagged: 0.589 at zero shots, a dip to 0.466 at one shot, the peak 0.648 at three shots, and 0.570 at five shots. At 31 B the curve is much smoother (0.664/0.748/0.780/0.676) and the one-shot dip has vanished entirely. At 125 B the same smooth pattern holds (0.748/0.787/0.858/0.787). The three-shot peak is therefore the right anchor at every scale, but the cost of moving away from it becomes more forgiving as the model grows.

The relative ranking of prompt templates changes with scale. At 8 B the minimal few-shot template wins, with the chain-of-verification variant lagging by 0.030. At 31 B that ranking flips: the chain-of-verification template becomes the best three-shot configuration (0.813 vs 0.780 for the minimal template). At 125 B it flips again, and the chain-of-verification template becomes the worst of the three (0.707 vs 0.858 for the other two). This is consistent with a simple emergent-versus-scaffolding story: at 31 B the model benefits from explicit list-check-flag-classify scaffolding because it cannot yet decompose the task internally, whereas at 125 B the same

scaffolding constrains a model that already chains its reasoning by default and forces it into a rigid output schema.

Two of the post-hoc levers we relied on at 8 B lose their power at the largest scale. Confidence-threshold tuning gave a small lift of 0.003 on the best 8 B configuration; on the best 125 B configuration the validation-optimal threshold is zero, meaning that every accepted-as-fake verdict is already worth accepting and any further filtering throws away signal. Self-consistency is similarly inverted in sign. At 8 B the voted score (0.597) is 0.051 below the greedy score (0.648). At 125 B that gap balloons to 0.182 (voted 0.676 vs greedy 0.858), and none of the five sampled runs at non-zero temperature even matches the greedy baseline. As scale grows, the greedy output is no longer noisy enough for averaging to help.

Scale dominates prompting on this task. The same three-shot frames prompt that gave 0.648 at 8 B reaches 0.858 at 125 B, a gain of more than 0.20 macro-F1, while the best individual prompting trick at 8 B (a bias-correction threshold sweep) was worth 0.003 on the same configuration. Whether this validation-set scaling story carries over to the held-out test split is a separate question, and as we will see in 4.5 the answer is less straightforward than the validation numbers suggest.

4.5 Final test-set evaluation

We now move to the held-out test split. Four System 2 configurations are locked from validation and each is evaluated on test exactly once: 8 B with the minimal three-shot frames prompt, 31 B with the same prompt for an apples-to-apples scaling row, 31 B with the chain-of-verification template (the validation-best three-shot template at 31 B), and 125 B with the minimal three-shot frames prompt (the validation-best configuration at 125 B). Table 5 reports macro-F1, accuracy, and per-class precision and recall for each row, alongside the System 1 baseline and the trivial classifier. The headline System 2 row is the 31 B model with the minimal three-shot frames prompt: it reaches a macro-F1 of 0.697 on test, above the TF-IDF baseline (0.627) and well above the trivial *always-real* floor (0.403).

The validation-to-test transfer is the most informative part of this table. Figure 1 plots validation versus test macro-F1 for each of the four System 2 rows. The configuration with the smallest gap (−0.083) and the best test score (0.697) is the simplest one in the comparison: 31 B with the minimal three-shot prompt. The configuration with the largest gap (−0.228) is the one that scored highest on validation: 125 B with the same prompt, which fell from 0.858 on val to 0.630 on test. The two remaining rows fit the same pattern. The chain-of-verification template at 31 B, the validation-best three-shot template at that scale, drops by 0.206 on test and ends up below the simpler 31 B configuration. 8 B with the same minimal three-shot prompt drops by 0.136 to 0.512. In short, configurations that scored highest on validation generalise worst to test, and the gap grows with both model scale and prompt elaboration.

The mechanism behind this pattern is consistent with the val-overfitting story we previewed at the end of 4.2. Validation has 59% fake samples; test has 32%. A configuration whose verdicts skew more toward *fake*, whether because of prompting tweaks or additional scale, is rewarded on validation and punished on test.

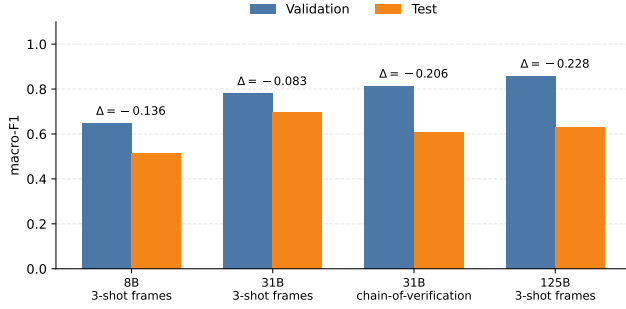


Figure 1: Validation versus test macro-F1 for the four System 2 configurations evaluated on the held-out test split. The transfer gap grows with both model scale and prompt elaboration; 31 B with the minimal three-shot prompt has the smallest gap and the best test score among the no-claims rows.

The validation-best rows in our grid are exactly the ones that over-predict fake on test: the 31 B chain-of-verification row flags 47 of 74 test samples as fake and the 125 B row flags 41, against a true count of 24. Their precision drops accordingly, and macro-F1 falls with it. The simpler 31 B three-shot row flags 40 samples as fake, the closest of any System 2 row to the test prior, and is also the only row whose recall on *fake* (0.875) is bought without sacrificing more than a small amount of precision.

The headline result of this report follows from table 5 directly: at 31 B parameters, a minimal three-shot multimodal prompt with four sampled video frames beats the TF-IDF baseline by 0.07 macro-F1 on the held-out test split. The val-best larger and more elaborated alternatives, which would have been the temptation to promote to the headline, all generalise less well than this simpler choice.

System	Macro-F1	Acc.	P _{fake}	R _{fake}	#fake
Trivial always- <i>real</i>	0.403	0.676	—	0.000	0
TF-IDF + LR (System 1)	0.627	0.635	0.462	0.750	39
8 B 3-shot frames	0.512	0.514	0.370	0.708	46
31 B 3-shot frames	0.697	0.703	0.525	0.875	40
31 B 3-shot CoV	0.608	0.608	0.447	0.875	47
125 B 3-shot frames	0.630	0.635	0.463	0.792	41

Table 5: Final test-set evaluation. All System 2 rows are locked from validation: greedy decoding, no threshold tuning on test. The trivial classifier and the TF-IDF baseline are included as reference rows. Bold marks the no-claims System 2 headline row.

4.6 Multi-stage claims pipeline

The claims pipeline of 3.4 tests whether distilling the noisy transcript and on-screen text into a structured list of verifiable claims helps the verifier focus on what should actually be checked. We ran the full pipeline on the validation split with two verifier models, 31 B and 125 B. With three prompt templates, minimal three-shot,

Template	31.3 B			125.1 B		
	no claims	+ claims	Δ	no claims	+ claims	Δ
Minimal three-shot	0.780	0.754	-0.026	0.858	0.688	-0.170
Role-priming	0.787	0.754	-0.033	0.858	0.676	-0.182
Chain-of-verification	0.813	0.664	-0.149	0.707	0.754	+0.047

Table 6: Validation macro-F1 with and without the claims block across two verifier models and three prompt templates. Five of six cells get worse when the extracted claims are added; the bold cell is the only one that gains.

role-priming and chain-of-verification, giving six model-template cells. We then locked two of those cells from validation and evaluated them once on test, mirroring the lock-from-val protocol used elsewhere in this report. Table 6 reports the validation grid; Table 7 reports the two locked test rows alongside the matching no-claims rows from 4.5.

On validation, five of the six cells get worse with claims, and three of them by a substantial margin. At 125 B the minimal three-shot template drops from 0.858 to 0.688 and the role-priming template drops from 0.858 to 0.676. At 31 B the chain-of-verification template drops from 0.813 to 0.664. The two remaining 31 B rows lose smaller amounts, -0.026 for the minimal template and -0.033 for role-priming. The single positive cell is 125 B with chain-of-verification, which moves from 0.707 to 0.754 and recovers toward the level of the simpler 125 B templates without claims. The cost of producing this validation grid was non-trivial: roughly two and a half hours of compute on the largest model for the claim-extraction stage, plus a further eleven hours for the six verifier runs.

Stage 1 itself produced clean output. The large extractor returned a parsable JSON list of claims for 88 % of its inputs, and an explicit empty list for many of the remaining cases that genuinely contained no verifiable claim; the residual 12 % of calls that produced unparseable output we treat as “no claims”, which is conservative. The claims that came back are recognisably factual in the form the prompt asked for: dates, places, named entities, statistics, and attributions, with very little of the surface style of the source text. The val penalty therefore appears at consumption time rather than at extraction time. Across the 197 videos that produced any claims at all, the mean count is 7.1 claims from the spoken transcript and 3.3 from the on-screen text, with several political-content videos producing fifteen or more claims in combination. A short claim list plausibly helps the verifier focus on what to check; a long one adds substantially more evidence-like text to the prompt than the post itself contained, and the verifier responds to that extra content by predicting fewer fakes overall.

The picture inverts on the held-out test split. We locked two configurations from validation: 31 B with the minimal three-shot template plus claims, the apples-to-apples claim-aware analogue of the 4.5 headline, and 125 B with chain-of-verification plus claims, the only val cell where claims gave a positive Δ over the matching no-claims row. The 31 B claim-aware row reaches a test macro-F1 of 0.758, a val-to-test gap of +0.004 and 0.061 above the no-claims test headline. This is the highest test score recorded by any System 2 configuration in this report. The 125 B claim-aware row reaches 0.608 on test, with a val-to-test gap of -0.146; the gap is

smaller than for no-claims at the same scale with -0.228 , but the absolute test score sits below the no-claims 125 B test row and the configuration also produces six unparsable responses out of seventy-four, suggesting that the structured chain-of-verification template combined with a long claim block stresses the JSON output enough to hurt parsing.

Configuration	Val	Test	Gap
31 B 3-shot frames, no claims	0.780	0.697	-0.083
31 B 3-shot frames + claims	0.754	0.758	+0.004
125 B 3-shot frames, no claims	0.858	0.630	-0.228
125 B 3-shot CoV + claims	0.754	0.608	-0.146

Table 7: Locked test rows for the claims pipeline alongside the matching no-claims rows from 4.5. Adding the claim block at 31 B closes the val-to-test gap to within noise and produces the highest System 2 test score in this report.

The simplest explanation of the val-test inversion is that the claim block makes the verifier slightly more conservative about flagging fakes, and that “slightly more conservative” lands close to the truth on val but much closer to the truth on test. At 31 B the no-claims row flags 19 of 29 val samples (66 %) and 40 of 74 test samples (54 %), against true fake counts of 17 (59 %) on val and 24 (32 %) on test. The claim-aware row flags 16 of 29 val samples (55 %) and 33 of 74 test samples (45 %). The claim-aware row is therefore one sample below the truth on val and nine samples above the truth on test, while the no-claims row is two samples above the truth on val and sixteen samples above on test. The val class balance happens to reward the no-claims row’s slight over-prediction; the test class balance punishes it, and the claim-aware row’s smaller fake-prediction count ends up closer to the held-out truth. This is the same val-overfitting story we documented in 4.5 for model size and prompt elaboration, applied along a different axis.

Taken together, the claims pipeline is not the unambiguous negative result the validation grid suggested. Adding extracted claims to the 31 B verifier’s prompt closes the val-to-test gap to within noise and gives the highest System 2 test score we report; at 125 B the same intervention narrows the val-to-test gap but does not overtake the simpler no-claims baseline. The hypothesis behind the design, that structured intermediate signal helps, survives in a modified form: the structured signal helps on test, even though providing it costs val macro-F1. The next obvious experiment, which we did not run, would feed the extracted claims into an external verification step rather than back into another prompted LLM, decoupling the claim extraction from the same fake-leaning prior that operates on the verifier. We return to this in the conclusion.

4.7 Error analysis on the headline configuration

Of the seventy-four held-out test samples, the headline System 2 configuration gets seventeen wrong, a 23 % error rate. The errors are still lopsided toward false positives, with thirteen real posts called fake against four fakes called real, but the imbalance is milder than in the no-claims headline of 4.5 with nineteen false positives and three false negatives. Compared sample by sample with that

no-claims headline, fourteen of the seventeen claim-aware errors are shared with no-claims, eight no-claims errors are recovered by claims, and three claim-aware errors are unique to this configuration. Confidence is also marginally better calibrated: of the thirty-seven predictions the model emits with self-reported confidence 1.0, four are wrong (11 %, against 18 % at the same self-reported threshold in the no-claims row). Most of the failures are stable across these two System 2 rows, which suggests that they reflect dataset-level failure modes rather than artefacts of either prompt.

The four false negatives all share one pattern. Each is a fake post built around a real topic, real-looking footage, or a real public figure, and the verifier treats topic-grounding as evidence that the specific claim of the post is also true. The “Tucker Carlson supports Democratic Socialism” post is flagged real because, in the model’s own words, “the video frames clearly show Tucker Carlson speaking on his network, and the on-screen captions align with the extracted claim from the transcript”. The “Fani Willis was indicted for indicting Trump” post is flagged real because the model recognises the underlying Fulton County legal proceedings and a Georgia case the post mentions, without noticing that the post inverts the direction of the actual case. Adding the claim block did not help in either example, and in the Tucker Carlson case the extracted claim is cited explicitly as evidence in the wrong direction.

Among the thirteen false positives, two clusters dominate. The larger cluster, accounting for roughly seven of the thirteen, is style-based false flag: the verifier uses formatting cues (sensationalist hashtags, “BREAKING” overlays, conspiratorial framing, hyperbolic language, merch-promo account framing) as evidence of fakeness. A representative example is a real Biden gaffe video flagged because the on-screen captions read “the brew beard” and the model reads the meme-style overlays as indicating “the video is a meme or edited clip rather than a real speech”. Another is a real-news compilation flagged because the on-screen text uses “highly subjective and hyperbolic language” while claiming to be unbiased. None of these rationales is unreasonable in isolation, since clickbait styling does correlate with misinformation in general, but the FakeTT annotation rule defines fakeness by the truth of the specific claim, not by stylistic markers, and the model’s strong style prior misfires accordingly.

The smaller cluster, three of the thirteen, is hallucinated fact-correction: the verifier identifies the post’s factual claim, contrasts it with what it believes is true of the world, and finds the post wanting on a contradiction the model itself has manufactured. The “egg in soda dissolves the shell” post is flagged because the model declares the chemistry “scientifically impossible”, although the phosphoric and citric acids in cola do dissolve calcium-carbonate shells over hours. The “invisibility blanket” post is flagged because the model declares the technology “scientifically impossible at this scale”, although the post is in fact a demonstration of an optical-illusion novelty. These are pure model-knowledge errors and persist unchanged from the no-claims row in 4.5; the claim block is no help when the verifier itself is producing the wrong counter-fact.

A third, smaller failure mode is unique to the claim-aware configuration. In two of the thirteen false positives, the verifier uses the *absence* of extracted claims as a signal of fakeness. A real instructional baseball-drill post is flagged because the rationale notes that “no verifiable factual claims were extracted from the video,

and the content is instructional/promotional rather than a factual report”; a real political-appreciation post about Donald Trump is flagged on similar grounds. Both videos genuinely contained no factual claims to extract, and the empty claim block was meant to be neutral, but the verifier appears to have acquired a small “empty claim list $\rightarrow$ suspicious” association from elsewhere in the prompt. This is a side-effect of the claims pipeline rather than a generic prompted-MLLM problem, and it cancels some of the precision gains the claim block delivers elsewhere.

Taken together, the system’s residual failures concentrate on a small number of patterns that the verifier cannot fix from within: style-cue confusion, hallucinated counter-facts, topic-grounding-as-evidence, and the new empty-claim suspicion introduced by the claims block itself. Of these, only the hallucinated fact-correction cluster has a clear external remedy, namely routing the extracted claims through a retrieval step against a known-fake or general-knowledge corpus, which remains the most plausible direction for closing the residual gap. We return to this in the conclusion.

4.8 Threats to validity

We close this section with a short list of factors that limit how confidently the test results above can be generalised.

Small test set. The test split contains seventy-four samples. Confidence intervals on macro-F1 are correspondingly wide: swapping two samples between the correctly-classified and the misclassified groups moves the metric by roughly 0.02 to 0.03, which is comparable to the gap between several adjacent rows in table 5. We report point estimates throughout but do not report bootstrap intervals, which a fuller evaluation would include.

Inverted class balance between validation and test. Validation is 59 % fake; test is 32 % fake. This single fact shapes the discussion throughout 4.5 and 4.6: configurations whose verdicts skew toward *fake* are advantaged on validation and disadvantaged on test regardless of whether their predictions are otherwise accurate. Any decision rule or prompt configuration selected on validation inherits this bias. We have tried to expose the bias rather than hide it through the lock-from-val protocol and the val-to-test gap analysis, but it remains a real constraint on what these results can claim.

Single random seed. Each pipeline configuration was run once with a fixed seed for the few-shot exemplar selection and other stochastic input-construction steps. Run-to-run variance from server-side decoding noise is not reported, except inside the self-consistency analysis, which itself was run on only one validation configuration per model size. A more rigorous evaluation would average metrics over multiple seeds and report confidence intervals.

Claim-extraction parse failures. Twelve percent of the stage-1 extraction calls returned output that our parser could not reduce to a structured list of claims, and we treat those as “no claims”. This choice is conservative for the metric, since it removes some of the claim block’s potential signal, but it understates the ceiling of the pipeline: a stricter re-prompt or a more tolerant parser would recover at least some of the lost claims. The headline test number for the claim-aware row should therefore be read as a lower bound on what the claims pipeline could achieve with the same prompts.

Remote inference dependency. All multimodal LLMs in this report are served by a remote inference cluster. During the model-scaling validation runs the server returned a small number of transient errors that caused two cells of the validation grid to be skipped: the self-consistency cell at 31 B, and the zero-shot text-only cell at 125 B. The headline test configurations were re-run cleanly and are not affected, but a fully reproducible run of the validation grid would require a more tolerant retry policy or a self-hosted inference setup.

5 CONCLUSION

We built two systems for fake-news detection on the FakeTT TikTok corpus: a TF-IDF and Logistic Regression text classifier as System 1, and a prompted multimodal LLM verifier with four sampled frames, post text, and balanced few-shot exemplars as System 2. On top of the ablation grid required by the brief, we ran a model size scaling study at three scales and a multistage claim extraction pipeline that distills each video’s transcript and on-screen text into a structured list of claims for the verifier to consume. On the held out test split, the best System 2 configuration reaches macro-F1 of 0.758, above the TF-IDF baseline at 0.627 and well above the trivial *always-real* floor at 0.403. The dominant pattern across the study was not capability but generalisation: configurations that scored highest on the fake-skewed validation split routinely lost more macro-F1 on the inverted-balance test split, with the largest validation best configuration losing 0.228 and the simplest validation-defensible alternative almost exactly matching its validation score on test. The lock from val protocol the brief mandates prevented us from chasing the validation best alternatives, and that discipline turned out to be the single most consequential design choice we made.

Limitations. The constraints discussed in 4.8 apply directly to these conclusions. The retained test split contains seventy-four samples, so individual rows of table 5 have wide implicit confidence intervals; multiple seeds and bootstrap resampling would tighten the comparisons. The validation split’s small size and inverted class balance also mean that any selection on validation, including the lock from val protocol used here, is itself a noisy procedure.

Future work: external claim verification. The claims pipeline as built distills each video into a structured list of factual statements but then routes those statements back to another prompted multimodal LLM, which inherits the same priors as the verifier. The error analysis in 4.7 showed that the residual failure clusters fall almost entirely outside the reach of the verifier: hallucinated fact corrections need an external knowledge source, and topic grounding errors need external claim verification rather than another internal pass. A natural extension is to feed the extracted claims to a retrieval step against a general knowledge corpus or a known fake database, decoupling the claim extraction from the same fake leaning prior that biases the verifier.

Future work: multi-agent verification. A second extension is to break the verifier itself into specialized agents, for example, one focused on visual evidence, one on textual style, and one on factuality that has a separate aggregator that resolves their disagreements. Most of the negative results in this report share a common structure in which the same prompted model family fills every role, so

the components systematically agree on the same wrong answers. A heterogeneous architecture could, in principle, make those disagreements visible and exploitable.

REFERENCES

- [1] Tom B. Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, Sandhini Agarwal, Ariel Herbert-Voss, Gretchen Krueger, Tom Henighan, Rewon Child, Aditya Ramesh, Daniel M. Ziegler, Jeffrey Wu, Clemens Winter, Christopher Hesse, Mark Chen, Eric Sigler, Mateusz Litwin, Scott Gray, Benjamin Chess, Jack Clark, Christopher Berner, Sam McCandlish, Alec Radford, Ilya Sutskever, and Dario Amodei. 2020. Language Models are Few-Shot Learners. arXiv:cs.CL/2005.14165 <https://arxiv.org/abs/2005.14165>
- [2] Christopher D. Manning, Prabhakar Raghavan, and Hinrich Schütze. 2008. *Introduction to Information Retrieval*. Cambridge University Press.
- [3] Soroush Vosoughi, Deb Roy, and Sinan Aral. 2018. The spread of true and false news online. *Science* 359, 6380 (2018), 1146–1151. <https://doi.org/10.1126/science.aap9559> arXiv:<https://www.science.org/doi/pdf/10.1126/science.aap9559>
- [4] Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc Le, Ed Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. 2023. Self-Consistency Improves Chain of Thought Reasoning in Language Models. arXiv:cs.CL/2203.11171 <https://arxiv.org/abs/2203.11171>
- [5] Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed Chi, Quoc Le, and Denny Zhou. 2023. Chain-of-Thought Prompting Elicits Reasoning in Large Language Models. arXiv:cs.CL/2201.11903 <https://arxiv.org/abs/2201.11903>
- [6] Xinyi Zhou and Reza Zafarani. 2020. A Survey of Fake News: Fundamental Theories, Detection Methods, and Opportunities. *Comput. Surveys* 53, 5 (2020), 1–40. <https://doi.org/10.1145/3395046>